DATABASE SYSTEMS ENGINEERING & DISTRIBUTED BACKEND DEVELOPMENT

INTELLIGENT EMAIL MANAGEMENT

AND SPAM DETECTION PLATFORM

PROJECT REVIEW — 2 • GROUP 12

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Project Problem & Objectives

Email volume grows.
Structure must keep up.

Spam introduces phishing, fraud and unwanted content. At scale, email records become difficult to manage, search and retrieve without a structured database.

PROJECT OBJECTIVES

Build the database foundation

Normalized relational schema • user and email information • spam-detection records • efficient search and retrieval

Indexing • joins • triggers • stored procedures

Literature Review: Research Areas

01

Email Storage

Efficient storage of email data

02

Email Management

Organization of large collections

03

Search & Retrieval

Efficient discovery of stored emails

04

Email Categorization

Automatic organization and classification

05

Spam Detection

Identification of spam emails

AIM / Understand existing approaches — then identify what can integrate into our DBMS project.

Selected Research Papers

Database-Driven Mail Servers Elprin & Parno	Database-based email storage
CEM — Conceptual Email Manager Cole & Stumme	Email organization & retrieval
Multi-Faceted Information Retrieval Perkiö et al.	Email search & analysis
Experimental Email Categorization Mock	Categorization & management
Review of Spam Email Detection Jáñez-Martino et al.	Spam detection & dataset shift

Email Storage & Management

PAPER 1 / Elprin & Parno

AN ANALYSIS OF DATABASE-DRIVEN MAIL SERVERS

Compared database-driven and traditional email-storage systems; studied MySQL for email storage. Evaluated searching and retrieving email information.

LIMITATION / Performance differs by database operation.

RELEVANCE / Supports MySQL for structured email storage and retrieval.

PAPER 2 / Cole & Stumme

CEM — A CONCEPTUAL EMAIL MANAGER

Shows that flexible organization improves the retrieval of large email collections.

LIMITATION / The conceptual system does not provide a relational DBMS implementation.

TAKEAWAY / Organization and database structure must be designed together.

Retrieval & Categorization

PAPER 3 / Perkiö et al.

MULTI-FACETED INFORMATION RETRIEVAL

Supports content-based search, topic analysis, social-network analysis and time-based analysis.

LIMITATION / Focuses mainly on information retrieval and analysis.

RELEVANCE / Supports efficient email search and retrieval.

PAPER 4 / Mock

EXPERIMENTAL FRAMEWORK FOR EMAIL CATEGORIZATION

Combined automatic categorization with retrieval to reduce manual organization.

LIMITATION / Not focused on a centralized, normalized relational database.

TAKEAWAY / Classification is stronger when backed by structured records.

Literature Review: Spam Detection

PAPER 5 / JÁÑEZ-MARTINO ET AL.

Spam patterns do not stand still.

The review examines detection techniques, spammer strategies, dataset changes and the dataset-shift problem.
Key finding: Changing spam patterns can reduce detection performance over time.

PROJECT RELEVANCE

Detection matters — but our project also manages email and spam information as durable DBMS records.

Comparative Analysis

DATABASE STORAGE	**Efficient email storage**	No complete spam management
EMAIL ORGANIZATION	**Flexible management**	Not focused on relational DBMS
INFORMATION RETRIEVAL	**Powerful search**	Centred on retrieval & analysis
CATEGORIZATION	**Automated organization**	Limited database focus
SPAM DETECTION	**Detection + evolution**	Mainly detection

OVERALL OBSERVATION / Research solves the parts — generally as separate systems.

Critical Analysis & Synthesis

DATABASE

Structured storage supports efficient retrieval.

MANAGEMENT

Organization, search and categorization improve usability.

SPAM

Detection matters — and patterns change over time.

SYNTHESIS / A practical email system must store, organize, search and manage email and spam records efficiently.

Research Gap

STORAGE + ORGANIZATION + RETRIEVAL + CATEGORIZATION + DETECTION

THE MISSING LAYER

ONE INTEGRATED RELATIONAL DBMS

Managing users, emails and spam-detection records with normalization, efficient retrieval and database-level operations.

Proposed Solution & Database Approach

1. USER MANAGEMENT
2. EMAIL MANAGEMENT
3. SPAM DETECTION MANAGEMENT
4. SEARCH & CLASSIFICATION
5. DATABASE TESTING

DBMS FEATURES / Normalized schema • primary & foreign keys • indexing • SQL queries • joins • triggers • stored procedures

RESEARCH FINDINGS → INTEGRATED DBMS SOLUTION

REFERENCE LINKS

1 https://www.researchgate.net/publication/221300666_An_Experimental_Framework_for_Email_Categorization_and_Management

2 https://dl.acm.org/doi/10.1145/3020165.3020175

3 https://dl.acm.org/doi/10.1145/3020165.3020175
https://link.springer.com/article/10.1007/s10462-022-10195-4?utm_source=chatgpt.com

4 https://dl.acm.org/doi/10.1145/3020165.3020175
https://link.springer.com/article/10.1007/s10462-022-10195-4?utm_source=chatgpt.com

5 https://www.researchgate.net/publication/221158654_Multi-Faceted_Information_Retrieval_System_for_Large_Scale_Email_Archives

THANK YOU